SmartNanotubes

Technical Specifications

Detector
Smell.iX16 – Type 5

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1. Purpose of the detector

The Smell.iX16 detector is designed to detect smells.

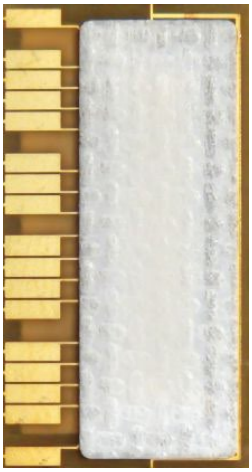
Product:	multichannel gas detector (Smell.iX16)
Type:	5
Version:	non-specific
Serial number:	unique, on-detector ID

2. Performance characteristics

Stability:	Insensitive to mechanical shocks
Measurement type:	Chemiresistor
Response range:	-80% – 10000% relative change on base resistance
Power consumption:	1 µW
Temperature:	-20°C to +70°C / -4 F to +158 F
Sensitivity:	High to different smells
Accuracy:	± 3% relative change of base resistance under exposure to artificial air, otherwise dependent on environmental conditions
Pressure range:	Atmospheric pressure ±10 %
Resistance:	low against water and any other liquid
Protection:	protected by IP67 filter against exposure to oil, polymer and water vapor

3. Layout

Material:	flexible polyimide foil, with particle filter
Dimensions:	23.0 x 12.3 mm / 0.91 x 0.47 inch (LxW)
Weight:	1 g / 0.002 lb
Design:	



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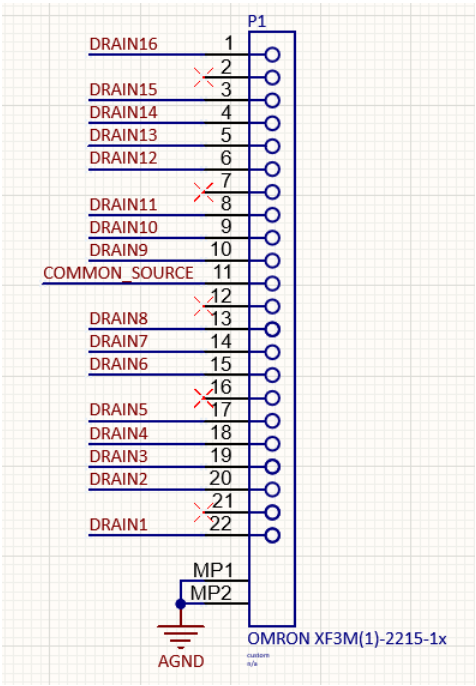
4. Detector requirements

Mounting: 22 pin ZIF socket 1mm pitch, detector for wire bonding on request
Conditions to avoid: Ozone concentration > 200 µg/m^3

ZIF socket pinout drawing:



5. Detector measurement requirements

Measure voltage: Measure with constant voltage on source electrode (supply voltage can be selected from a range of 50 mV - 120 mV)
Measuring range: 0,2 kΩ to 10 MΩ
Measurement accuracy: Recommended

- < 10 kΩ: 10 Ω
- >= 10 kΩ: 0,1 %

Sampling interval: Recommended

- 2 – 10 s for each channel

6. Producer

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7. Disclaimer

- **Caution:** Information provided without warranty, subject to change without notice. Further details available through sales department, latest version on the SmartNanotubes Technology website.
- **Packaging:** Recycling recommended, return of sorted material possible upon agreement (costs borne by the customer). Additional charges for unsorted returns.
- **Safety / Medical Applications:** Not tested for critical applications. Customers must inform SmartNanotubes Technologies before planned use for further coordination.

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